

Research Project: Franciscan Eco-Theology and Environmental Attitudes

A term paper submitted for the
RM in Management and Applied Microeconomics

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Abstract

Our project focuses on the effect of the exposure to the Franciscan and Dominican orders between the 12th and 15th centuries on contemporary environmental attitudes. We find that that higher Franciscan exposure is associated with less pro-environmental attitudes, while higher Dominican exposure is associated with more pro-environmental attitudes contrary to our initial expectations. We also notice that the effect of the exposure to the two orders could be driven by the question on collective action, which is likely the underlying mechanism which is consistent with the contrasting theologies of the two orders regarding collectivity and individualism.

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1 Introduction

Religious institutions, in particular the Church, played a pervasive role in every aspect of life in medieval Europe, shaping people’s lives, values, and behavior throughout the Middle Ages (Lynch, 2014). One of the most famous religious figures of the Medieval Ages is St. Francis of Assisi, who is the founder of Franciscan order. Being recognized by the Catholic Church as the patron saint of ecology, he is widely known for his deep affection towards nature and animals.

In this research, we study the Franciscan eco-theology and its effect on contemporary environmental attitudes. We have also controlled for the Dominican order, which is the other famous Mendicant order of the Medieval Ages with comparably equal historical importance. Thus, our research question is the following: What is the effect of exposure to the Franciscan "Eco-Theology" from the 1200s to the 1500s on contemporary environmental attitudes among European populations? (compared to the Dominican theology).

To do so, we first construct our independent variable - the historical exposure measures for both Franciscan and Dominican orders, defined as a fraction of a region’s historical population living geographically close to a monastic house. We then construct our dependent variable - environmental attitudes index - by applying Principal Component Analysis to the individual-level survey data on environmental attitudes from the European Values Study (EVS) 2017–2021 dataset. Using those variables and the set of controls, we empirically examine how historical exposure to the two Mendicant orders affects contemporary environmental attitudes.

Given the historical background of St. Francis of Assisi, we have set the following initial hypothesis: we expected that higher historical exposure to the Franciscan Order is associated with more pro-environmental attitudes, while higher historical exposure to the Dominican Order is associated with less pro-environmental attitudes. However, our empirical results show the opposite. Contrary to our initial expectations, we find that higher historical exposure to the Franciscan Order is associated with less pro-environmental attitudes, while higher historical exposure to the Dominican Order is associated with more pro-environmental attitudes.

2 Historical Context and Hypothesis

The Franciscan Order was founded in the early 13th century by St. Francis of Assisi (1181-1226), who was recognized by the Catholic Church as the patron saint of ecology, widely known for his profound affection towards nature and animals. This deep affection towards animals and nature can be seen in his "Canticles of the Creatures," where St. Francis refers to the sun and the moon as brother and sister. Importantly, Francis and his followers emphasized the close imitation of life of Christ and apostles through the ideals of poverty and humility. "He wanted to embrace real poverty, without any rationalisations or modifications. His band of followers would beg for their daily food, would take no care for tomorrow, and would own nothing – really nothing." (Lynch, 2014, p. 258). Thus, the Franciscan Order focuses on collective life, communal care, antimaterialism, shame, compassion, simplicity, and humbleness (Arrunada and Lopez-Manuel, 2024).

In contrast, the Dominican Order, founded by St. Dominic (1170-1221) in the early 13th century, prioritized intellectual and educational mission of medieval Catholic Church.

They founded and staffed universities and produced extensive theological and practical writings (Le Goff, 1980; Boyle 1981). The Dominicans emphasized the importance of critical thinking, self-examination, and moral development (Hinnebusch 1966; Lesnick 1989).

Because of the strong association of the Franciscan Order with nature and animals, we would naturally expect more-environmental attitudes in regions with stronger Franciscan exposure. Therefore, we initially hypothesized that higher historical exposure to the Franciscan Order would be associated with more pro-environmental attitudes, while higher historical exposure to the Dominican Order would be associated with less pro-environmental attitudes. However, our final results state the opposite.

3 Literature Review

A similar study was conducted by Arruñada and López-Manuel (2024), who examined the effects of the two main Mendicant orders of the Late Middle Ages - the Dominican and the Franciscan. Through their emphasis on guilt, shame, compassion, and analytical thinking, these orders shape outcomes at the cognitive, interpersonal, and institutional levels, which together constitute the “foundations” of impersonal exchange. They further argue that eco-theological beliefs can foster distinct values and cultural norms. For example, at the interpersonal level, “Dominicans promoted guilt as a driver of prosocial behavior toward strangers”, fostering a culture of individual accountability and independence, “whereas Franciscans encouraged compassion and shame, which strengthened bonds within the group” (Arruñada and López-Manuel, 2024). Consistent with these differences, the authors document contrasting effects of Dominican and Franciscan influence on impersonal exchange. Thus, their findings supports our results for distinct effects of the two orders on environmental attitudes.

Our study also relates to the broader literature on the role of Medieval Churches and persistence of cultural norms. The Church, as an institution, played a pervasive role in every aspect of life in medieval Europe, shaping people’s lives, values, and behavior throughout the Middle Ages (Lynch, 2024). Once a religious institution becomes established, the values and norms it historically imposes tend to persist, as such cultural legacies “leave an imprint on values that endures” over time (Inglehart and Baker, 2000). Although religious institutions have lost some of the formal allegiance of their followers in modern societies, Inglehart and Baker’s findings suggest that religiously rooted value orientations continue to influence individuals, as these values become embedded in national cultures and are transmitted through educational institutions and mass media, allowing religiously rooted value orientations to persist at the individual level.

Another important strand of literature relates to the determinants of environmental attitudes. In many literatures it was examined that cultural differences might influence environmental attitudes (Chwialkowska et al., 2020; Khan et al., 2024; Yang et al., 2024; Tam, 2025). Some literature also suggests that Religiously grounded moral worldview of nature is associated with how people understand and relate to the environment (Cohen et al., 1976). Some theoretical models argue that pro-environmental attitudes may be rooted in self-interest and utility maximization. In particular, a longer life expectancy may increase individuals’ willingness to support environmental efforts to fight climate

change (Mariani et al., 2010). Additionally, importantly for our research, many other literature with similar research interests commonly measure the environmental attitudes using the survey datasets (EVS) (Gelissen, 2007; Behler, 2025).

Our research is the first to examine the effect of the exposure to Franciscan and Dominican monasteries on contemporary environmental attitudes.

4 Data and Methodology

4.1 Exposure Measure

In our research, we assume that people that are living close to the monasteries can get influenced or exposed to the theological beliefs of the monasteries. Thus, the exposure measure is defined as a fraction of a region's historical population living geographically close to a monastic house. To estimate historical exposure to the Franciscan and Dominican orders, we borrow from a lot of datasets. First, we employ the geodataset covering the years between 1216 to 1500 from the Mapping Past Societies (MAPS), which includes information on the geographic locations, the latitude and longitude, of the Dominican monasteries consisting of 870 observations and Franciscan monasteries consisting of 994 observations and the names of the houses. The Dominican dataset includes the foundation centuries. The only limitation of this dataset is that it does not include the ending years. After we get the monastery location, we need to locate them within regions. To do so, we use the European Commission system for characterising geographical data, the Nomenclature of Territorial Units for Statistics (NUTS), as well as country shapefiles. NUTS-1 are for large macro regions, like German states, NUTS-2 are for more localized regions within countries. Since Germany has modified their NUTS administrative borders, we rely only on the larger NUTS-1 regions for Germany, and use NUTS-2 for other countries. We needed to delete monasteries in east Ukraine, Crimea, and Middle Eastern cities (Jerusalem/Acre), since those monasteries lie in the locations that are not in or close to any NUTS region. There are some monasteries that are a little bit outside but still close to NUTS regions, so we coerce them to fit into the closest regions, such as into Romania, Poland or Greece.

Once we have combined the monastery locations with NUTS and shapefile, our next step is to observe historical population data.

HYDE 3.3 (History Database of the Global Environment) Utrecht University population data captures where people lived historically. It is an estimate of population count and density on regular latitude-longitude grid from the very late Medieval time to the modern period. Grids are very small squares of 85 km^2 , and we have an estimate of population density for each grid as seen in Figure 2(a).

Similarly, we also consider the population not only in squares, but also in the NUTS regions. By doing similar calculations, we can get the exposure measure for each NUTS-2 region as seen in Figure 2(b).

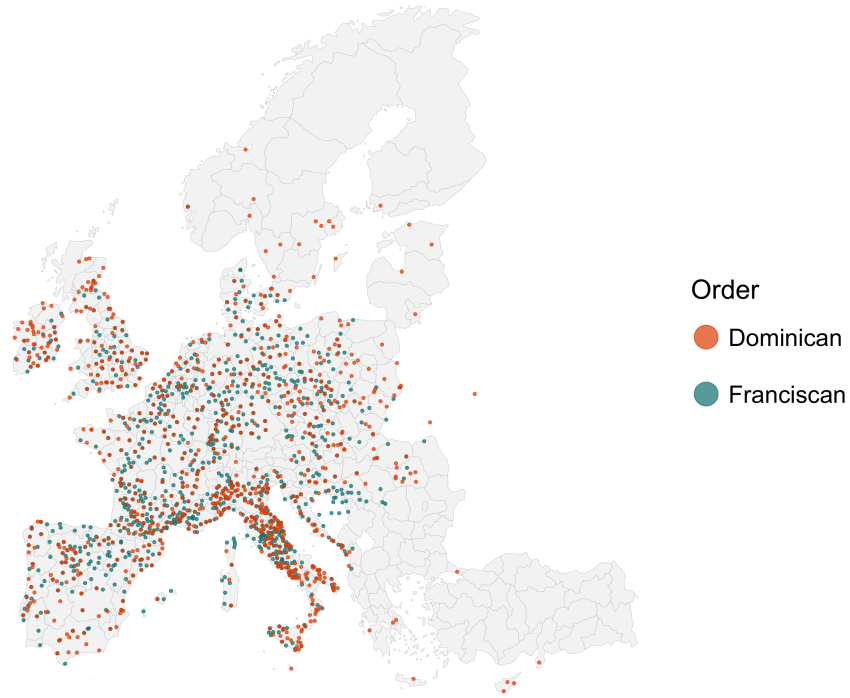
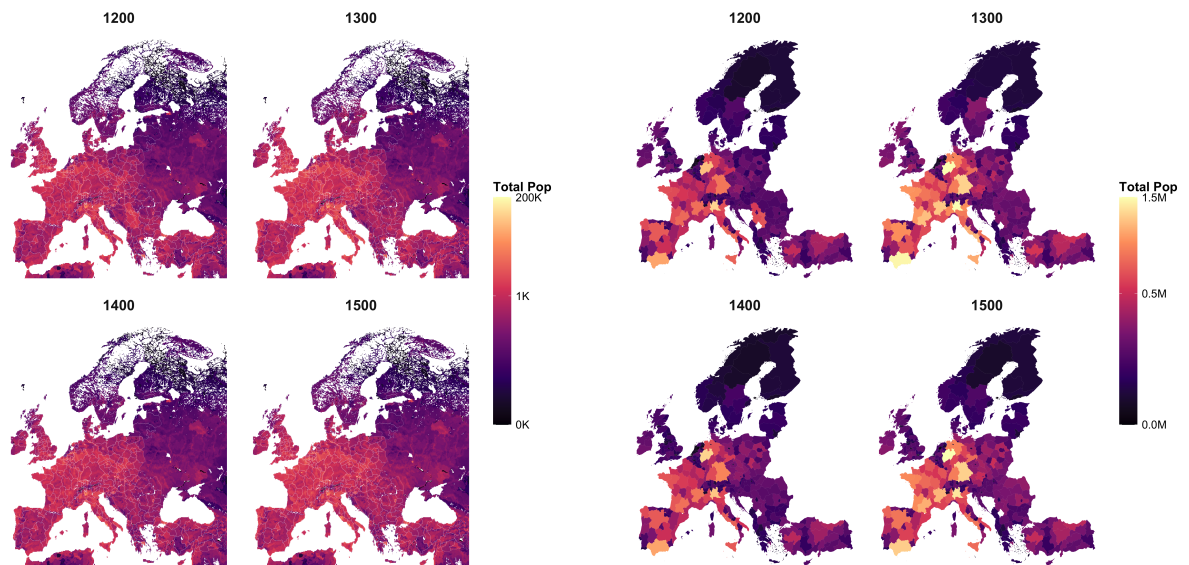


Figure 1: Locations of Franciscan and Dominican Monasteries



(a) Population densities (grid-level, log scale)

(b) Regional populations (NUTS-2 level)

Figure 2: Population measures from HYDE 3.3

We draw a circle around the monastery, and all the little squares (grids) that are inside that circle are counted as the exposed population. All squares that are mostly included in the circle are counted as fully exposed to the theology of that monastery. In our research, the exposure measure is defined as the sum of exposed population within a given region as a percentage of the total population for both Franciscan and Dominican monasteries:

$$\text{Exposure} = \frac{\text{Exposed Population}}{\text{Total Population}}$$

Having all the three sources put together, we have Dominican and Franciscan exposure measure for Europe (Figures 3a and 3b).

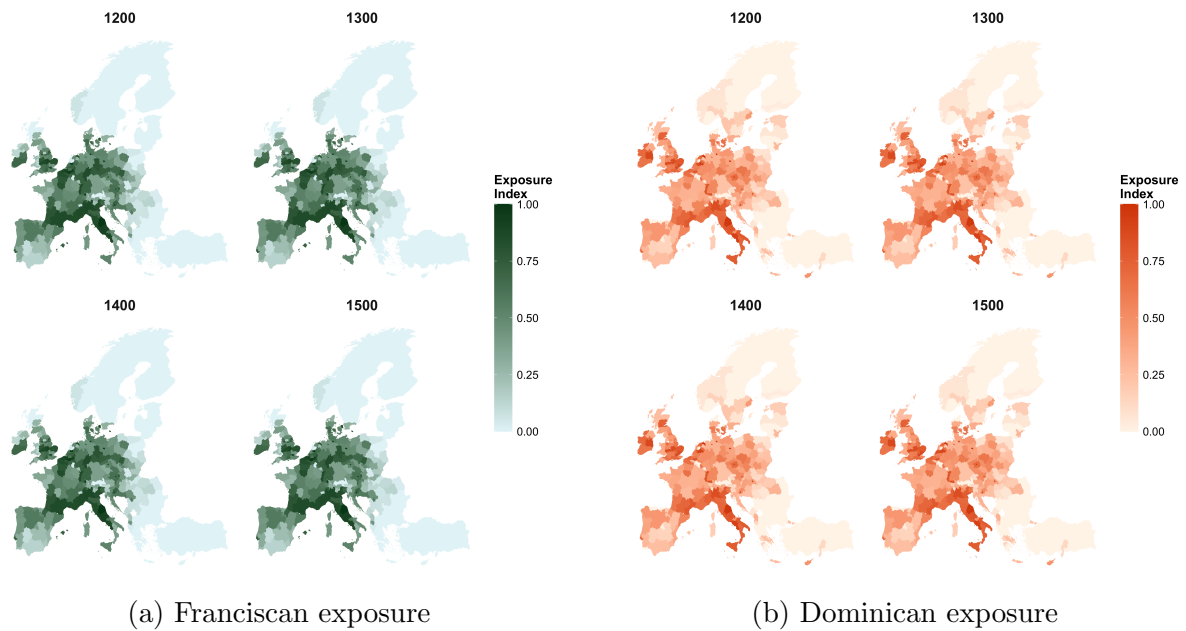


Figure 3: Map of exposure to Mendicant Orders in Europe (1200–1500)
normalized index (0 = none, 1 = high exposure)

4.2 Environmental Attitudes Measure

To construct our main dependent variable, the environmental index, we use the European Values Study (EVS) dataset 2017–2021, which is individual-level survey data on values, beliefs, attitudes, and socioeconomic characteristics across European countries. The dataset covers over 30 European countries of which 24 are relevant for our research when we exclude the countries that are not in NUTS-2. After the data cleaning, the dataset includes 49233 observations and geographic identifiers for NUTS-1 regions for Germany and NUTS-2 for all other countries.

From the EVS dataset, we borrow particularly 6 environmental variables, which are the questions that are related to environmental attitudes. The questions are the following:

- Choosing either "to protect the environment, even if slower economic growth and loss of jobs" or "to have economic growth and newly created jobs, even if environment suffers". - Economy and Environment Trade-off question
- "I would give part of my income if I were certain that the money would be used to prevent pollution."
- "It is just too difficult for someone like me to do much about the environment."
- "There are more important things to do in life than protecting the environment."
- "There is no point in doing what I can for the environment unless others do the same." - Collective action question
- "Many of the claims about environmental threats are exaggerated."

The questions have categorical answers ("strongly agree", "agree", "neither agree nor disagree", "disagree", "strongly disagree", or choosing between the two statements). However, the questions differ in their directional interpretation. Thus, we need to standardize the questions by assigning 0-1 scale per answer for each question, such that the answers indicate the same environmental direction. In the standardized 0-1 scale, 1 indicates the least pro-environmental attitude, and 0 indicates the most pro-environmental attitude. See Figure 4 for visualization for environmental questions within NUTs 2 regions. The green regions are more pro-environmental, while the yellow regions are less pro-environmental.

Since we have 6 different closely related questions for environmental attitudes, we want to reduce the dimensionality of the data and construct a single Environmental Attitudes Index. Thus, we run a Principal Component Analysis (PCA) to try to extract the main components of the questions, invisible or underlying mechanisms that are driving the answers. The first principal component explains 45% of the variance, and we take the first principal component as our Environmental Attitudes Index. We find that the components of Western European countries behave more green, while the Eastern European countries behave more yellowish as illustrated in Figure 5.

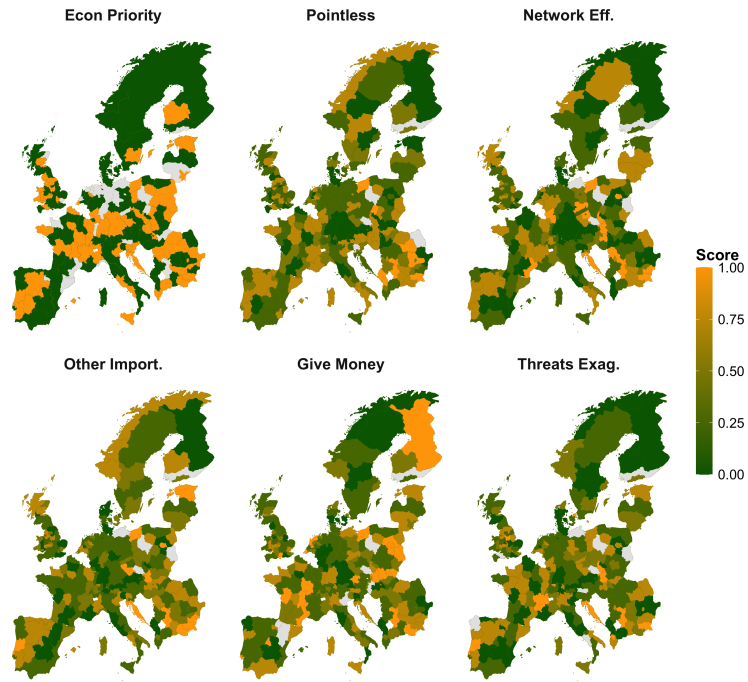


Figure 4: Distribution of the 6 environmental questions across Europe considering the average scores per region.

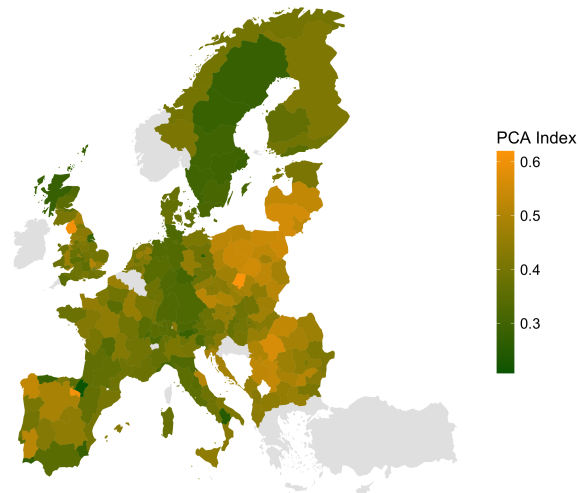


Figure 5: Environmental Index (PCA) across Europe considering the average scores per region.

4.3 Regression Model

Once we have all the variables put together, we run our regression:

$$Y_{irc} = \alpha + \beta_1 Franc_r + \beta_2 Dom_r + X'_i \gamma + \delta_c + \varepsilon_{irc} \quad (1)$$

where Y_{irc} denotes the environmental attitude index for individual i in region r of country c (scaled 0-1), $Franc_r$ and Dom_r represent population-weighted exposure to Franciscan and Dominican orders in region r measured in 1500, and X'_i is a vector of control variables, δ_c denotes country fixed effects, and ε_{irc} is the error term.

The control variables were also taken from the EVS data, the same dataset as the environmental question variables. The vector of individual-level control variables, X'_i , consists of gender, age, education, household income (PPP-adjusted), socio-economic status (ISEI), town size, political orientation (left–right scale), and religious denomination with dummy variables for Catholic and Protestant. We cluster standard errors at the NUTS level.

5 Results

5.1 Regressions and Robustness Checks

We started by running the first four regressions shown in Table 1 and found the following results. Model 1 is the simplest form of the regression, where we regressed the environmental index on the Franciscan exposure only. And we see that the coefficient β_1 is negative and statistically significant. Because the environmental index ranges from 0 to 1, where lower values indicate greater pro-environmental behavior and higher values indicate less pro-environmental behavior, a negative coefficient indicates that the variable makes people more pro-environmental. β_1 in Model 1 was in accordance with our initial hypothesis that higher Franciscan exposure would make people more pro-environmental. In Model 2, adding the Dominican exposure decreases β_1 . When we add country fixed effects in Model 3, β_1 changes its sign and loses its significance. In Model 4, when we finally add all the individual-level controls on top of the Dominican exposure and country fixed effects, β_1 becomes fully positive and statistically significant, which means that regions that had maximum Franciscan exposure had a 3.5 percentage points lower environmental index. This is quite an interesting result for us. We could also see that many of the control variables have a significant effect. For example, females are more pro-environmental. However, the coefficient of the Dominican exposure, β_2 , is not statistically significant in any of the models we have analysed. Even though we cannot say much about it, β_2 is always negative, indicating that generally the Dominican exposure is correlated with a more pro-environmental attitude.

Table 1: Regression Results (SE clustered by region)

	Model 1 Franciscans	Model 2 Fran vs Dom	Model 3 Country FE	Model 4 Full + FE
Franciscan Exposure	−0.095*** (0.016)	−0.058* (0.027)	0.004 (0.014)	0.035* (0.015)
Dominican Exposure	—	−0.050 (0.027)	−0.013 (0.014)	−0.016 (0.014)
Female	—	—	—	−0.028*** (0.003)
Age	—	—	—	0.000** (0.000)
Education	—	—	—	−0.000*** (0.000)
Income (PPP)	—	—	—	−0.002 (0.001)
Right-wing	—	—	—	0.007*** (0.001)
Town Size	—	—	—	−0.005** (0.001)
Socio-Economic Index	—	—	—	−0.001*** (0.000)
Catholic	—	—	—	0.004 (0.004)
Protestant	—	—	—	−0.001 (0.005)
Observations	49,233	49,233	49,233	24,379
R^2	0.022	0.024	0.099	0.149
Adj. R^2	0.022	0.024	0.099	0.148

Notes: Standard errors in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

All the β_1 and β_2 coefficients from these regressions are plotted in Figure 6. β_1 has a steady increase going from a negative to a positive sign as we move from Model 1 to Model 4, and finally becomes positive and statistically significant. β_2 is never significant; however, it also decreases once we add controls.

In Table 2, we run regressions with each of the environmental questions as dependent variables separately to see whether there is one variable within the PCA index driving all of the results. We run the regression in Model 4 six times with one of the variables instead of the index. The values of β_1 and β_2 did not change that much across these six regressions compared to Model 4 in Table 1. However, there aren't any significant results for them except the variable indicating the network and collectivity effect, stating, "There is no point in doing what I can for the environment unless others do the same." The Franciscan and Dominican exposure coefficients produce significant results since they have contrasting theologies regarding this variable, where the Franciscan order focused more on collectivity and the Dominican order emphasized an individualistic view.

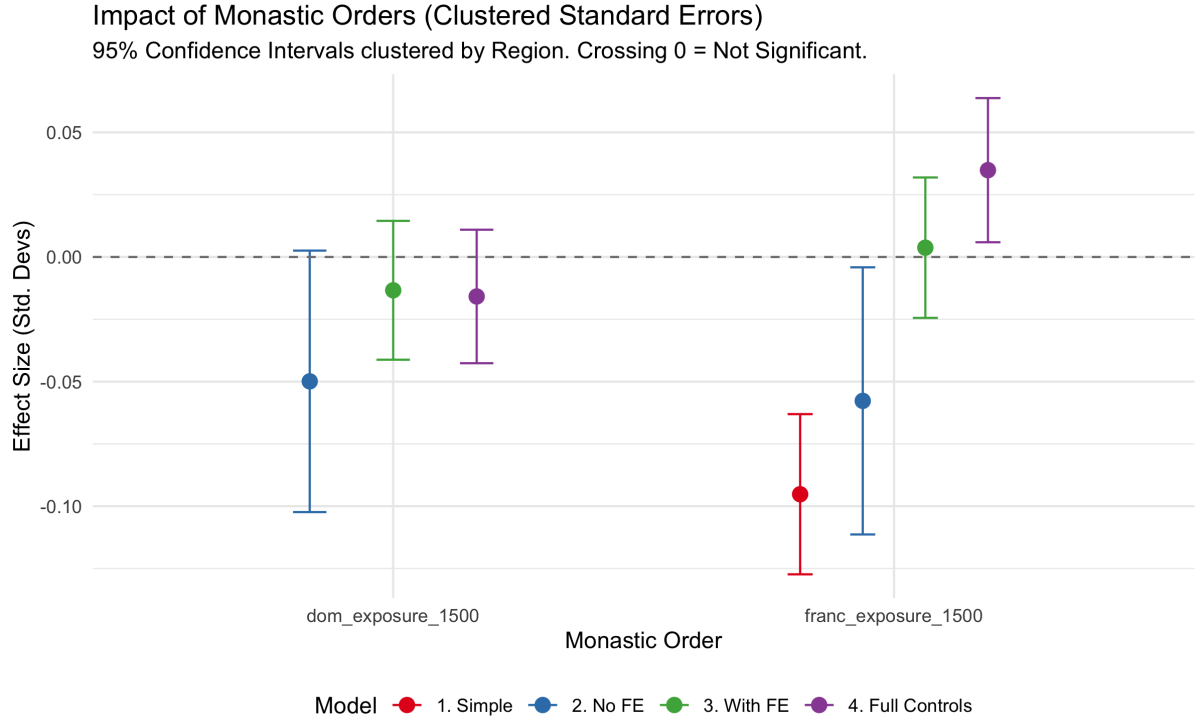


Figure 6: Impact of Monastic Orders (Clustered SEs)
95% CI clustered by region. Crossing 0 = Not Significant

In Figure 7, we run the main regression in Model 4 twenty-four times, each time excluding one country at a time to see whether there is one country driving the results. The plot on top shows the Dominican exposure coefficient, β_2 , estimates across these regressions, and the bottom plot shows the Franciscan exposure coefficient, β_1 , estimates. We see that there isn't much difference across countries. Only Switzerland and Italy have a slightly stronger effect on the results, but even excluding these countries, the results remain quite similar to those in Table 1. So the effects are homogeneous across Europe.

Table 2: Individual Environmental Questions (Clustered SEs)

	(1) Econ Priority	(2) Give Money	(3) Too difficult	(4) Other Import.	(5) Collective	(6) Threats Exag.
Franciscan Exp.	0.052 (0.038)	0.016 (0.021)	0.034 (0.021)	0.030 (0.019)	0.064** (0.021)	0.037 (0.021)
Dominican Exp.	-0.020 (0.037)	-0.026 (0.021)	-0.014 (0.018)	-0.018 (0.016)	-0.039* (0.016)	-0.017 (0.015)
Gender (Female)	-0.027*** (0.007)	-0.002 (0.004)	-0.023*** (0.004)	-0.040*** (0.004)	-0.028*** (0.005)	-0.047*** (0.004)
Age	0.001** (0.000)	0.001*** (0.000)	0.002*** (0.000)	0.000* (0.000)	0.001*** (0.000)	0.001*** (0.000)
Education	-0.000** (0.000)	-0.000*** (0.000)	-0.000*** (0.000)	-0.000*** (0.000)	-0.000*** (0.000)	-0.000*** (0.000)
Income (PPP)	-0.002 (0.002)	-0.004* (0.002)	-0.011*** (0.002)	-0.004** (0.001)	-0.010*** (0.002)	-0.004** (0.001)
Socio-Econ. Ind.	-0.002*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)
Town Size	-0.003 (0.004)	-0.006** (0.002)	-0.001 (0.002)	-0.004* (0.002)	-0.009*** (0.002)	-0.012*** (0.002)
Polit. Orient. (R)	0.011*** (0.003)	0.006*** (0.002)	0.003** (0.001)	0.004*** (0.001)	0.005*** (0.001)	0.010*** (0.002)
Catholic	0.007 (0.009)	0.005 (0.007)	0.011* (0.006)	0.010 (0.005)	0.017* (0.007)	0.007 (0.007)
Protestant	-0.003 (0.013)	0.003 (0.012)	-0.007 (0.006)	-0.017** (0.005)	-0.004 (0.006)	0.008 (0.008)
Num. Obs.	25,427	27,212	27,344	27,370	27,343	26,823
R^2	0.095	0.121	0.140	0.141	0.125	0.121
Adj. R^2	0.094	0.120	0.139	0.139	0.124	0.120

Notes: Standard errors in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

In Table 3, we run the main regression again, this time excluding each control variable at a time for the sensitivity analysis. We see that the estimate of β_1 remains around 0.035 and is statistically significant across most of the regressions, except for one regression in which we exclude the town size control variable, which accounts for urbanization. This is an important result because the Franciscans used to locate themselves within cities and urban areas to reach more people. So, not controlling for the town size would bias the coefficient of Franciscan exposure. This makes the town size a very important control, but apart from it, we could drop any other control and find very similar results to our full model.

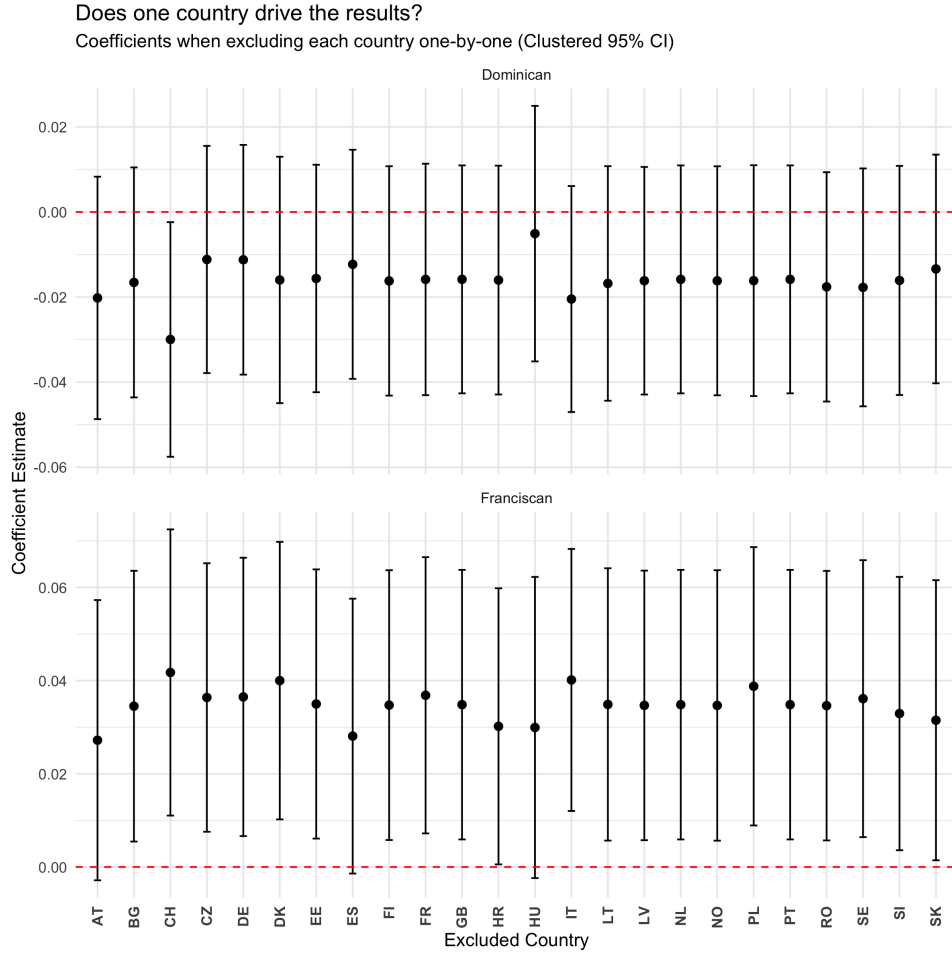


Figure 7: Does one country drive the results?
Coefficients when excluding each country one at a time (Clustered 95% CI)

Table 3: Sensitivity Analysis: Coefficients when Dropping Controls

	Full Model	Excl. Gender	Excl. Age	Excl. Education	Excl. Income	Excl. Politics	Excl. Town Size	Excl. Socio- Econ. Index	Excl. Religion
Franc. Exp.	0.035*	0.036*	0.035*	0.035*	0.032*	0.033*	0.012	0.039**	0.034*
Dom. Exp.	-0.016	-0.016	-0.015	-0.017	-0.008	-0.010	-0.010	-0.020	-0.016
Num. Obs.	24,379	24,379	24,434	24,438	26,795	33,560	28,220	27,286	24,379
R^2	0.149	0.144	0.149	0.147	0.143	0.135	0.141	0.148	0.149
Adj. R^2	0.148	0.143	0.147	0.145	0.142	0.134	0.139	0.147	0.148

Notes: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

In Figure 8, we see the plotted regression coefficient estimates found in the sensitivity analysis in Table 3 by leaving out one control in each regression, the top plot showing estimates for β_2 and the bottom one showing estimates of β_1 . We can clearly see that only the town size shifts the Franciscan exposure coefficient estimate. The Dominican exposure coefficient estimate also changes slightly when we remove income, political orientation, and town size, but none of the controls change the general direction of the results.

As the last robustness check in Table 4, we run subsample regressions with the full model, including all controls, but keeping only Catholic individuals in the first subsample. The Catholicism of individuals is determined from the same EVS data by asking the question “Which religious denomination do you belong to?” and taking into account only those who answered with the option “Roman Catholic”. Considering only Catholics, the sample size drops significantly to 5899, so we don’t have any significant results and can’t say much about them. In the second subsample, we exclude the observations from Germany to only keep NUTS 2 regions to see whether it was the lack of linearity in Germany driving the results, since we consider NUTS 1 regions for Germany. But we see that the results stay pretty much the same as in the main model.

Table 4: Robustness Checks: Subsamples

	Main Model (All)	Catholics Only	Exclude Germany
Franciscan Exposure	0.035* (0.015)	0.023 (0.021)	0.037* (0.015)
Dominican Exposure	-0.016 (0.014)	0.004 (0.018)	-0.011 (0.014)
Num. Obs.	24,379	5,899	23,182
R^2	0.149	0.132	0.146
Adj. R^2	0.148	0.126	0.145

Notes: Standard errors in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

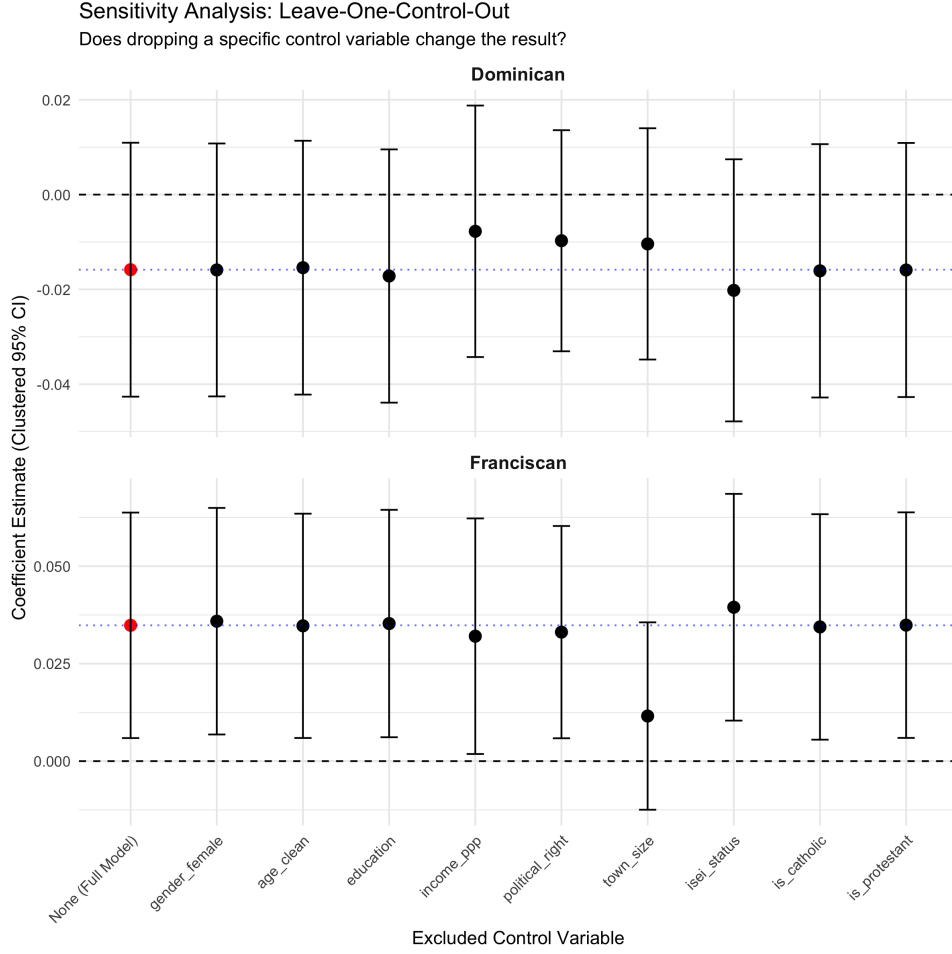


Figure 8: Sensitivity Analysis: Leave-One-Control-Out
Does dropping a specific control variable change the results?

5.2 Possible Mechanism

Since we found contrasting results to what we were expecting to find, we didn't want to leave it like this and want to propose a possible mechanism that might be driving these results, which could be used for further hypotheses and research. We have already ruled out all misleading mechanisms that could have come from the control variables, such as education, income, and political views, and from country fixed effects. For example, one could have claimed that Dominican exposure also meant a higher education level since they promoted teaching, which could have led to more pro-environmental attitudes. However, the mechanism should be a deep, slow-moving, cultural or institutional effect.

We propose that the mechanism could lie in the opposite beliefs about taking collective action within the Franciscan versus Dominican theologies. The Franciscan theology is the theology of humility and acceptance of circumstances, poverty, and scarcity for the sake of imitating Christ. They value moral purity over institutional action. The lack of desire for collective intervention might have shown itself through having less pro-environmental attitudes, which could also be observed in our variables, beliefs about individual efforts being pointless unless others do the same, and environmental threats being exaggerated. The Franciscans are not likely anti-nature but rather anti-intervention. The Dominican theology, on the other hand, is the theology of persuasion and institutional coordina-

tion. Their willingness to take reasoned collective action and enforce norms could have translated into more modern pro-environmental attitudes. Their sense of obligation also possibly played a role in the context of the environment, and they might have had greater trust in institutions for collective solutions against environmental threats.

6 Conclusion

7 Limitations and Further Research

8 References

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